Fiscal Year:	FY 2014	**Task Last Updated:**	FY 09/25/2014
PI Name:	Hargens, Alan R. Ph.D.		
Project Title:	Rodent spine deconditioning after 30 days of microgravity		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology (2) Neurobiology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Zip Code:	92037-0863	**Congressional District:**	52
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2007 Space Flight: Bion-M1 NNH07ZTT001N
Start Date:	08/01/2009	**End Date:**	07/31/2014
No. of Post Docs:	3	**No. of PhD Degrees:**	
No. of PhD Candidates:	4	**No. of Master' Degrees:**	2
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No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA ARC
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Flight Program:	Bion/Shuttle		
Flight Assignment:	STS-131 ; Bion M1 NOTE: Extended to 7/31/2014 per A. Chu/ARC (Ed., 4/30/2014)		
Key Personnel Changes/Previous PI:	Jan 2012. New Co-I: Brandon R Macias, PhD, NSBRI Postdoctoral Fellow, Department of Orthopaedic Surgery, University of California, San Diego, San Diego, CA 92103-8894, Email: b1macias@ucsd.edu ; Feb 2012. New Co-I: Jamila Siamwala, PhD, Postdoctoral Fellow, Department of Orthopaedic Surgery, University of California, San Diego, San Diego, CA 92103-8894.		
COI Name (Institution):	Lotz, Jeffrey (University of California, San Francisco) Masuda, Koichi (University of California, San Diego) Rana, Brinda (UCSD Cancer Center Sequencing Core) Sah, Robert (UCSD Bioengineering) Macias, Brandon (University of California, San Diego) Siamwala, Jamila (University of California, San Diego)		
Grant/Contract No.:	NNX09AP11G		
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Performance Goal Text:	
Task Description:	The vertebral bodies and flexible intervertebral discs of the spine are important, weight-bearing tissues that are adapted to gravitational stress. The absence of gravitational axial loads on the body during exposure to microgravity may disrupt normal spine physiology and function. During microgravity exposure, deconditioning of the intervertebral discs and spinal tissues poses a serious risk upon re-exposure to 1-G and exercise. Our recent data in identical twins exposed to 30-days bed rest indicate that compressibility increases in the lumbar spine and its intervertebral discs during 30-days simulated microgravity. Because intervertebral discs become more collapsible during simulated microgravity, the probability of disc prolapse or herniation upon re-exposure to gravity increases. We hypothesize that nucleus pulposus swelling pressures in mouse spines will significantly decrease following 15- and 30-days of microgravity. In addition, we hypothesize that lack of gravitational loading on lumbar discs during microgravity will lead to extracellular matrix damage and significantly reduced disc compression performance following 15- and 30-days of microgravity. To test our hypotheses, we will harvest and test spines from ground-based control mice and flight mice exposed to 15 and 30-days of actual microgravity on STS-131 and Bion-M1. Spines will be frozen and transported to UCSD and UCSF for accepted methods of testing nucleus pulposus swelling pressure, lumbar disc dimensions, proteoglycan content, and spine biomechanics. We will compare the results of the present study to our peer-reviewed and published results for identical twins exposed to 30-days simulated microgravity and control mice exposed to 1-G. Our research will aid understanding of spinal deconditioning during 15- and 30-days microgravity and of the higher incidence of disc prolapse or herniation following re-exposure to 1-G with a long-term view to prevent such spinal deconditioning with exercise or other physiologic countermeasures. Separately, evidence suggests headward fluid shifts alone influence bone alteration. However, most conclusions are drawn from human bed rest studies and hindlimb unloading of rats aimed to simulate actual weightlessness. As a member of NASA's Biospecimen Sharing Program, our lab acquired murine calvariae from a 15-day mission aboard Space Shuttle STS-131. In concordance with current theory, we hypothesized that compared to ground control mice, headward fluid shifts in mice exposed to microgravity will induce adaptive growth in bones that normally do not bear weight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	To our knowledge, this study is the first to examine the effects of 15- and 30-days space flight on spinal deconditioning in mice. The vertebral bodies and flexible intervertebral discs are important, weight-bearing tissues that have adapted to gravitational stress. Our research will aid understanding of spinal deconditioning during 15- and 30-days microgravity and of the higher incidence of disc prolapse or herniation following re-exposure to 1-G with a long-term view to prevent such spinal deconditioning with exercise or other physiologic countermeasures. This research may aid understanding of spinal deconditioning during inactivity such as after spinal cord injury and bed rest in human patients on Earth. Separately, our studies of mouse skull calvaria demonstrate the influence of the fluid shift associated with microgravity on bone remodeling independent of skeletal loading. The findings may be considered in the development of strategies to counter bone loss in astronauts and patients confined to bed rest.
	1) Research Products: We have used state-of-the-art technologies to quantify morphology, biochemistry, and kinematics of spines (including the vertebrae, intervertebral discs, spinal cord, and spinal muscles) of mice after 15 and 30 days of actual microgravity. Our research group is uniquely positioned to compare these data from animal models to that of changes that occur in crew members following 6-month ISS missions (NASA Awards NNX10AM18G and NNX13AM89G "Risk of Intervertebral Disc Damage after Prolonged Space Flight"). 2) Type of Investigation: Space Flight 3) Relevance of the Research Products to NASA: We completed a well-documented and validated battery of histological, biochemical, and biomechanical measures, developed and validated to identify and quantify spinal features for mice during space flight. We hypothesized that disuse and subsequent decrements in spinal muscle structure and function lead to alterations in biomechanical and biological pathomechanisms. First, microgravity leads to supraphysiologic disc swelling and increased disc height that may stiffen the lumbar motion segment and cause abnormal segmental movement patterns. These biomechanical changes may increase risk for disc degradation, vertebral endplate microfracture, and facet joint capsule strain. Second, swollen and unloaded discs in microgravity may have reduced nuclear matrix production and nutrient transport from endplate capillaries in adjacent vertebra. We propose that these factors may adversely affect disc cell metabolism, inducing disc matrix degradation (via decreased proteoglycan (PG) synthesis and increased secretion of matrix metalloproteinases). Moreover, the impaired nutrient transport may adversely affect vertebral bone homeostasis. 4) Map to Human Research Program Integrated Research plan: Our project directly addresses Critical Path Roadmap Risks and Questions regarding disc injury (IRP Gap-B4): Is damage to joint structure, intervertebral discs, or ligaments incurred during or following microgravity exposure? Our research will improve understanding of the underlying pathophysiology of spinal deconditioning induced by microgravity, and mechanisms of spinal adaptation following re-exposure to 1-G. Our long-term goal is to prevent such spinal deconditioning with exercise or other physiologic countermeasures. The goal of this research is to comprehensively characterize 15 and 30 days space-flight and recovery induced changes in spinal tissue morphology, biochemistry, and biomechanics of mice. 5) Productivity of NASA-Funded Research: Our research team has two NASA grants to study spines and their adaptations to actual microgravity: one grant to test mice exposed to 15 and 30 days of actual space flight (STS-131 and Bion, respectively) and another grant to test 12 ISS crew members before and after 180 days of actual space flight. In addition, our group was selected as part of a tissue sharing opportunity to initiate a new NASA project to study spines and calvaria from rats exposed to 90-days hind-limb suspension. Thus, from a scientific standpoint, it is highly desirable to compare results from the present 15 and 30 days of actual microgravity to those changes that occur in ISS crew members following approximately 180 days of actual microgravity. It is known that prolonged exposure to microgravity causes degeneration of the human spine, indicated by low back pain during space flight and increased incidence of post-space flight herniated nucleus pulposus. We examined the effect that microgravity poses on biomechanical properties of lumbar and caudal discs from mice on 15 day shuttle mission STS-131. Sixteen mice (space flight group,

n=8; control group, n=8) were sacrificed immediately after space flight. For caudal discs, the space flight group exhibited a 32% (p=0.012) lower physiologic disc height and 70% (p=0.02) lower strain dependence compared to the control mice. Decreases in both disc height and strain dependence indicate disc degeneration and injurious effect of microgravity. By contrast, differences were not noted for lumbar discs. This potentially deleterious interaction between prolonged weightlessness and differential range of motion along the spine may underlie the increased cervical versus lumbar disc herniation rates observed among astronauts (Johnston et al., 2010).

The increased herniation risk is highest during the first post-space flight year, and gradually subsides thereafter (Johnston et al., 2010). Consequently, we also hypothesized that the biomechanical properties of the intervertebral disc deteriorate during space flight but then recover after acclimation to normal gravity. To test this hypothesis, we compared the compressive creep properties of caudal discs after return from a 13-day Shuttle mission (STS-133) to those of ground-based control mice. Six flight and ten control mice were sacrificed +1 day, +5 days, or +7 days after the landing of STS-133. Compared to controls, space flight mice had 12% lower disc height and 21% lower strain-dependence on swelling pressure. In addition, biomechanical properties did not recover significantly over the 7-day post-flight period. Overall, biomechanical properties of the caudal discs were diminished by space flight, consistent with observations that prolonged exposure to microgravity increases disc herniation risk. These properties did not recover after short-term (7-days) re-acclimation to 1-G loading. Our separate studies of skull bones documented significantly increased growth and elastic modulus of skull bones following 15-days of microgravity exposure (Zhang et al., 2013).

Space flight Decreases Bending Strength and Alters Failure Mode in Murine Spinal Segments. Introduction: The effects of microgravity during space flight negatively impact the human spine and lead to increased risk for low back pain and injury. Intervertebral disc herniation risk is increased 4.3-fold following space flight [1]. Animal models are often used to study the effects of space flight on the spine, and compressive creep properties have been shown to decrease in mice post-space flight [2]. However, uniaxial compression alone does not cause disc herniation. Rather, hyper-flexion is a significant contributor to herniation, i.e. during heavy lifting in a flexed posture [3]. Due to the observed increase in herniation risk following space flight, we hypothesized that the bending strength of discs is decreased under microgravity conditions. The effects of space flight on spine bending properties have not previously been studied.

The specific aims of the current study were to determine: (1) mechanical bending properties; and (2) locations of bending failure in murine spinal segments following space flight. We hypothesized that biomechanical bending properties of post-flight mouse discs would be diminished and that specimens would fail within the annulus fibrosus or at the annular insertion to the vertebra.

Materials and Methods: The 14 animals used for this study were obtained through NASA's Biospecimen Sharing Program (BSP). Six mice were sent on a 30-day microgravity Bion M1 mission and acclimated to 1-G for 12 hours before they were sacrificed, while eight mice remained on the ground for the 30 days in standard vivarium cages. BSP personnel sacrificed mice and dissected and flash froze tissues. We received isolated frozen tails from the six post-space flight mice (flight, n=6) and eight ground vivarium controls (control, n=8).

Task Progress:

The C3-C4 motion segment was isolated from each tail. The ventral side of each motion segment was marked with tissue dye to track orientation. Surrounding ligaments and soft tissues were removed to isolate the discs and vertebrae, and each specimen was radiographed to estimate disc height and cross-sectional area. Specimens were kept hydrated with phosphate buffered saline throughout tissue preparation. Each motion segment was placed in a custom 3D-printed four-point bending device such that the plane of bending coincided with the ventral-dorsal plane. This bending device had stainless steel rods inserted through the top and bottom pieces, perpendicular to the plane of bending, to create the four contact points. The lower supports of the bending system were spaced 7.5 mm apart, while the upper supports were spaced 2.5 mm apart and attached to the moving cross-head of a materials test system (ElectroForce 3200; Bose, Eden Prairie, MN). Specimens were preloaded between 0 and 5 mm at a rate of 0.05 mm/sec for 5 cycles, and then loaded to failure at 0.006 mm/sec. Ramping was terminated after failure, characterized by a drop in force of 1.5 N. From force-displacement data, the toe region was defined as the displacement corresponding to an applied force of 1 N, the stiffness was defined as the slope of the most linear region, and the strength was defined as the maximum load before failure. Paired t-tests were used to assess differences in failure strength, failure displacement, toe region length, and disc height between the control and flight groups.

Results: Space flight reduced bending strength by 17% (6.27 ± 0.94 vs. 7.53 ± 1.13 N, $p < 0.05$) and reduced toe region by 32% (0.53 ± 0.13 vs. 0.77 ± 0.21 mm, $p < 0.05$) with no significant effect on bending stiffness ($p > 0.25$). Disc height was reduced by 16% in flight mice (0.25 ± 0.03 vs. 0.29 ± 0.04 mm, $p = 0.054$).

The two modes of failure identified by histology were: separation within the growth plate; and annulus avulsion at the disc-vertebra junction. Post-flight mouse spinal segments tended to fail within the growth plate while control spinal segments tended to fail at the disc-vertebra junction. Polarized light microscopy showed annulus collagen fibers inserting into the cartilage endplate but not continuing through the endplate to the subchondral bone.

Discussion: The reduction in bending strength following space flight supports our hypothesis that reduced bending properties may be an important factor for explaining microgravity-associated increased herniation risk. The reduction in toe region is consistent with reports of decreased range of motion in astronauts, and the shortened disc height is consistent with prior studies investigating post-flight mice [2]. All specimens failed at soft-hard tissue interfaces between cartilage and bone, presumably due to stress concentrations caused by abrupt changes in material properties at these junctions. The increased likelihood of post-flight mice to fail at the growth plate before the disc-vertebra junction suggests a decrease in growth plate integrity, and may explain the reduction in bending strength without a corresponding reduction in stiffness.

The results of this study motivate future work investigating the underlying biochemical and structural mechanisms of growth plate failure in flight mice. Previous work suggests that chondrocyte polarity and TGF-ß family signaling are essential for growth plate health [4,5]. A thorough investigation of these traits in post-flight mice may illuminate effects of microgravity that compromise murine growth plate integrity.

A limitation of animal models used in shuttle missions is the lack of clarity on whether post-flight changes are caused by the same mechanisms as in human astronauts. Microgravity likely has different effects on murine spines due to anatomical and physiological differences. The growth plate separations observed in this study are not relevant to adult humans, but may provide insight for mechanisms of vertebral growth plate fractures observed in children before growth plate closure. Interestingly, mice have been observed using their tails to move about inflight enclosures [2], which may

	be a confounding variable in accelerating disc degeneration. It should also be noted that the flight mice in this study acclimated to 1-G for 12 hours before they were sacrificed, and the effects on tissue properties during this period are unknown. Despite these limitations, this study provides the first analysis of bending properties of intervertebral discs following space flight and provides insight into possible mechanisms for increased herniation risk following space flight. Significance: Intervertebral disc herniation risk is increased over 4-fold following space flight. Here we show the first evidence that bending properties of spinal segments are decreased following space flight, which may contribute to increased herniation risk. References: [1] Johnston SL+, ASEM 2010; [2] Bailey JF+, GSB 2012; [3] Adams MA+, Spine 1982; [4] Abad V+, Endocrinology 1999; [5] Zhang J+, Developmental Biol. 2005 Over the past few years of NASA funding, we have published 8 peer-reviewed papers and 18 abstracts. Peer-reviewed publications are listed below. Bailey JF, Hargens AR, Cheng KK, Lotz JC. Post-spaceflight recovery of biomechanical properties of murine intervertebral discs. Gravitational and Space Biology 2012 Oct; 26(2):38-47. Bailey JF; Hargens AR; Cheng KK; Lotz JC. Effect of microgravity on the biomechanical properties of lumbar and caudal intervertebral discs in mice. Journal of Biomechanics, 47: 2983-2988, 2014. Hargens AR, Bhattacharya R, Schneider SM. Space physiology VI: exercise, artificial gravity, and countermeasure development for prolonged space flight. Eur J Appl Physiol 2013 Sep; 113(9):2183-92. Kim SH, Neuschwander TB, Macias BR, Bachman L Jr, Hargens AR. Upper extremity hemodynamics and sensation with backpack loads. Appl Ergon 2013 Sep 26. [Epub ahead of print] PubMed Rodríguez-Soto AE, Jaworski R, Jensen A, Niederberger B, Hargens AR, Frank LR, Kelly KR, Ward SR. Effect of load carriage on lumbar spine kinematics. Spine 2013 Jun 1; 38(13):E783-91. Sayson JV, Lotz J, Parazynski S, Hargens AR. Back pain in space and post-flight spine injury: Mechanisms and countermeasure development. Acta Astronautica 2013; 86:24-38. Shymon SJ, Yaszay B, Dwek JR, Proudfoot JA, Donohue M and Hargens AR. Altered disc compression in children with idiopathic low back pain: an upright magnetic resonance imagery backpack study. Spine 39: 243-248, 2014. Zhang B, Cory E, Bhattacharya R, Sah R, Hargens AR. Fifteen days of microgravity causes growth in calvaria of mice. Bone 2013 Oct; 56(2):290-5. We thank NASA for research support from Grant NNX09AP11G and for the outstanding assistance of NASA ARC's Biospecimen Sharing Program Team.
Bibliography Type:	Description: (Last Updated: 08/25/2025)
Articles in Peer-reviewed Journals	Bailey JF, Hargens AR, Cheng KK, Lotz JC. "Effect of microgravity on the biomechanical properties of lumbar and caudal intervertebral discs in mice." Journal of Biomechanics. 2014 Sep 22;47(12):2983-8. Epub 2014 Jul 14. http://dx.doi.org/10.1016/j.jbiomech.2014.07.005 ; PubMed PMID: 25085756 , Sep-2014
Articles in Peer-reviewed Journals	Kim SH, Neuschwander TB, Macias BR, Bachman L Jr, Hargens AR. "Upper extremity hemodynamics and sensation with backpack loads." Appl Ergon. 2014 May;45(3):608-12. Epub 2013 Sep 26. http://dx.doi.org/10.1016/j.apergo.2013.08.005 ; PubMed PMID: 24075289 , May-2014
Articles in Peer-reviewed Journals	Shymon SJ, Yaszay B, Dwek JR, Proudfoot JA, Donohue M, Hargens AR. "Altered disc compression in children with idiopathic low back pain: an upright magnetic resonance imaging backpack study" Spine (Phila Pa 1976). 2014 Feb 1;39(3):243-8. http://dx.doi.org/10.1097/BRS.0000000000000114 ; PubMed PMID: 24253789; PubMed Central PMCID: PMC3947616 , Feb-2014
Articles in Peer-reviewed Journals	Berg-Johansen B, Liebenberg EC, Li A, Macias BR, Hargens AR, Lotz JC. "Spaceflight-induced bone loss alters failure mode and reduces bending strength in murine spinal segments." J Orthop Res. 2015 Aug 18. [Epub ahead of print] http://dx.doi.org/10.1002/jor.23029 ; PubMed PMID: 26285046 , Aug-2015
Articles in Peer-reviewed Journals	Hargens AR, Vico L. "Long-duration bed rest as an analog to microgravity." J Appl Physiol (1985). 2016 Apr 15;120(8):891-903. Epub 2016 Feb 18. Review. http://dx.doi.org/10.1152/japplphysiol.00935.2015 ; PubMed PMID: 26893033 , Apr-2016
Articles in Peer-reviewed Journals	Macaulay TR, Siamwala JH, Hargens AR, Macias BR. "Thirty days of spaceflight does not alter murine calvariae structure despite increased Sost expression." Bone Rep. 2017 Dec;7:57-62. eCollection 2017 Dec. Epub 2017 Aug 18. https://doi.org/10.1016/j.bonr.2017.08.004 ; PubMed PMID: 28875158; PubMed Central PMCID: PMC5574818 , Dec-2017
Articles in Peer-reviewed Journals	Siamwala JH, Macias BR, Healey R, Bennett B, Hargens AR. "Spaceflight-associated vascular remodeling and gene expression in mouse calvaria." Front Physiol. 2022 May 13;13:893025. https://doi.org/10.3389/fphys.2022.893025 ; PMID: 35634164; PMCID: PMC9139491 , May-2022
Awards	Hargens AR. "Nello Pace Award for Outstanding Leadership in Gravitational Physiology, June 2014." Jun-2014
Awards	Hargens AR. "Citation Award from the American College of Sports Medicine, July 2014." Jul-2014